

Unsupervised Analysis Lab — PCA and Hierarchical Clustering

Outline — overview

M2ISDD — Bioactive Molecules

2026-09-10

RNA and protein binding pockets (real, anonymized data). Detailed working document: 2026-09-09_TP_ACP_CAH_QSAR_etudiant_EN.qmd. Packages: `FactoMineR`, `cluster`, `fpc`, `pheatmap`.

Session 1 — PCA

Active variables without `Rg` (to be computed by you): `rEV21`, `rEV32`, `rEV31`, `Num_Atoms`.

1. Getting familiar with the data (import, structure, missing values).
2. PCA on the 3 shape ratios alone, **without scaling** (`scale.unit = FALSE`), `Origin` as a supplementary variable. Spot a sign of redundancy between variables from the eigenvalues, then verify it numerically.
3. Compute `Rg = sqrt(Eigval1+Eigval2+Eigval3)`. PCA on 3 ratios + `Num_Atoms` + `Rg`, **without scaling**: spot an artifact (very heterogeneous raw variance), verify it numerically. Redo the same PCA **with** scaling (`scale.unit = TRUE`, the default): the structure changes completely.
4. Summary: do pocket shape and pocket size carry the same biological information? What role did scaling play in this session?

Session 2 — Hierarchical clustering, linkage criterion

Working space: the 3 shape ratios alone (`rEV21`, `rEV32`, `rEV31`), as in the source article. `Num_Atoms` and `Rg` are reserved for session 4.

1. Reload the data.
2. `hclust()` with 4 criteria: `single`, `complete`, `average`, `ward.D2`.
3. Compare the 4 dendrograms (overall shape, without discussing partitions yet).

4. Cut at $k = 4$ (provisional). Compare cluster sizes across criteria, then the ARI between criteria (`fpc::cluster.stats()$corrected.rand`).
5. Summary: which criterion would you keep, and why?

Session 3 — Choosing k and bootstrap stability

Start from the `ward.D2` clustering (or the criterion chosen in session 2).

1. Reload the data, rebuild the clustering.
2. Average silhouette for $k = 2$ to 8 (`cluster::silhouette()`), on the default scale **then** on the full, theoretical scale $[-1, 1]$ (with reference lines at 0.25/0.5) — compare what each scale shows you. Which k looks optimal? Is that score high?
3. Stability bootstrap **coded by hand** with `sample()`: resample, re-cluster, match clusters by maximum Jaccard index, average over B replicates. Apply to $k = 4$.
4. Repeat for $k = 2$ and $k = 3$, compare.
5. Summary: do silhouette and stability answer the same question?

Session 4 — Clustering $\times$ PCA, heatmap, comparing two clusterings

Same data as in sessions 1 to 3 (no new file — the complete file with `Archetype` is only used in the personal-reflection section at the end of the document).

1. Reload the data. Rebuild, exactly as before, the clustering on the 3 shape ratios alone, **without scaling**, as built in sessions 2-3 (not a new analysis). Then build a second clustering, a new one this time, on `rEV21`, `rEV32`, `rEV31`, `Num_Atoms`, and `Rg` (`Rg` to be recomputed), **with** scaling this time (necessary as soon as `Num_Atoms` enters the space, cf. session 1). Along the way: `Num_Atoms` and `Rg` are correlated ($r = 0.83$, session 1) — what consequence does that have on the effective weight of size in this clustering?
2. Rebuild PCA1 (shape) and PCA2 (shape+size) from session 1. Project `Origin` onto PCA2. On PCA1, project successively the clusters from the 5-variable clustering, then those from the 3-ratio clustering: does the size structure (5 variables) leave a visible trace in the pure-shape plane, unlike the shape-only structure (3 ratios)?
3. Heatmap (`pheatmap`) of the 5 variables, annotated with `Origin`, the cluster from the 5-variable clustering, and the cluster from the 3-ratio clustering. Do the color blocks of the two clustering annotations line up visually with the row reordering? Then check numerically: contingency table between the two clusterings, per-cluster Jaccard index, ARI.
4. Final summary: connect the 4 sessions to one another.

Going further — personal reflection (outside session time)

Outside lab time: load the complete data file (`../data/2026-09-09_data_PROT_RNA_TP_seance4_complet.csv`) which contains an additional **Archetype** column — this is not a clustering result, but a median-threshold rule defined in the source article. Compare your two session-4 clusterings to **Archetype**, then read the Results section of the article and confront it with your own results: archetype frequencies, shape-vs-size effect sizes, bootstrap stability of the threshold rule compared to the stability (or lack thereof) you obtained in session 3.